The vaxsurvR pipeline

A visual walk through the workflow, from raw export to coverage estimates

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Everything in this document is computed from `vcs_example`, the synthetic survey shipped with the package: 600 households, 256 children, 14 antigens, with a documented set of deliberately planted data-quality problems. No real participant data appears anywhere. Every figure is produced by the function named above it, so each page can be reproduced in one line.

1 The shape of the workflow

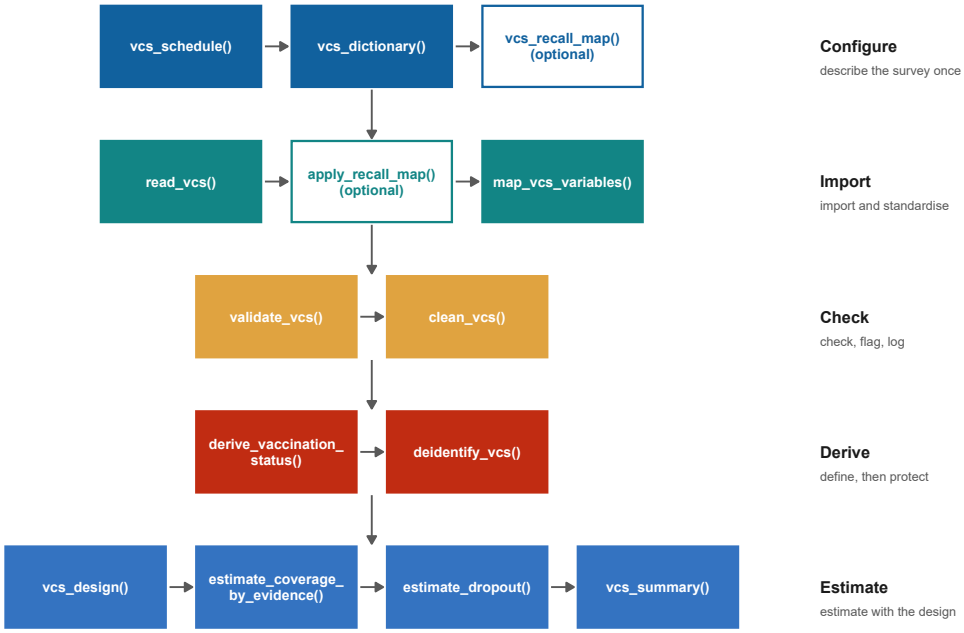


Figure 1: The five stages: plot_vcs_pipeline()

Read it top to bottom. The first row is the part people most often skip: **the survey is described once**, in a schedule and a dictionary, and every later call reads those two objects. That is what makes the same script run against next round's export without editing a single function.

The two outlined boxes are optional. A survey whose questionnaire has no caregiver-recall block simply omits them.

Table 1: The pipeline as a table: vcs_pipeline()

Stage	Step	Required	What it does
Configure	vcs_schedule()	required	describe the survey once
Configure	vcs_dictionary()	required	describe the survey once
Configure	vcs_recall_map()	optional	describe the survey once
Import	read_vcs()	required	import and standardise

Stage	Step	Required	What it does
Import	apply_recall_map()	optional	import and standardise
Import	map_vcs_variables()	required	import and standardise
Check	validate_vcs()	required	check, flag, log
Check	clean_vcs()	required	check, flag, log
Derive	derive_vaccination_status()	required	define, then protect
Derive	deidentify_vcs()	required	define, then protect
Estimate	vcs_design()	required	estimate with the design
Estimate	estimate_coverage_by_evidence()	required	estimate with the design
Estimate	estimate_dropout()	required	estimate with the design
Estimate	vcs_summary()	required	estimate with the design

2 Configure: describe the survey once

The **schedule** says which antigens exist, the earliest age each dose may validly be given, which dose precedes it and the minimum interval between them. Nothing in the package assumes a country's calendar; these are the numbers every later check reads.

Table 2: A vaccination schedule: vcs_schedule()

Vaccine	Antigen	Dose	Min age (d)	Max age (d)	After	Min interval (d)
BCG	BCG	1	0	28	NA	NA
OPV0	OPV	0	0	14	NA	NA
OPV1	OPV	1	42	76	OPV0	28
PENTA1	PENTA	1	42	76	NA	NA
PCV1	PCV	1	42	76	NA	NA
OPV2	OPV	2	70	104	OPV1	28
PENTA2	PENTA	2	70	104	PENTA1	28
PCV2	PCV	2	70	104	PCV1	28

The **dictionary** maps the export's own variable names onto standardised concepts. Wide exports encode repeat groups in the column name, so the dictionary takes templates: `{c}` for the caregiver repeat, `{k}` for the child, `{vv}` for the zero-padded vaccine index.

Table 3: Mapping source variables onto concepts: vcs_dictionary()

Concept	Source variable	Level
interview_id	KEY	household
household_id	KEY	household
psu	psu_id	household
segment	segment_id	household
stratum	stratum	household
province	province	household
district	district	household
health_zone	health_zone	household
health_area	health_area	household
residence	residence	household

3 Import: one object, three tables

`map_vcs_variables()` expands the repeat groups and returns a single object holding the household, child and vaccination views side by side, with the raw import carried along untouched.

Table 4: What the mapped object contains: map_vcs_variables()

Level	N
households	600
children	256
psus	30
segments	2
strata	2
vaccines	14

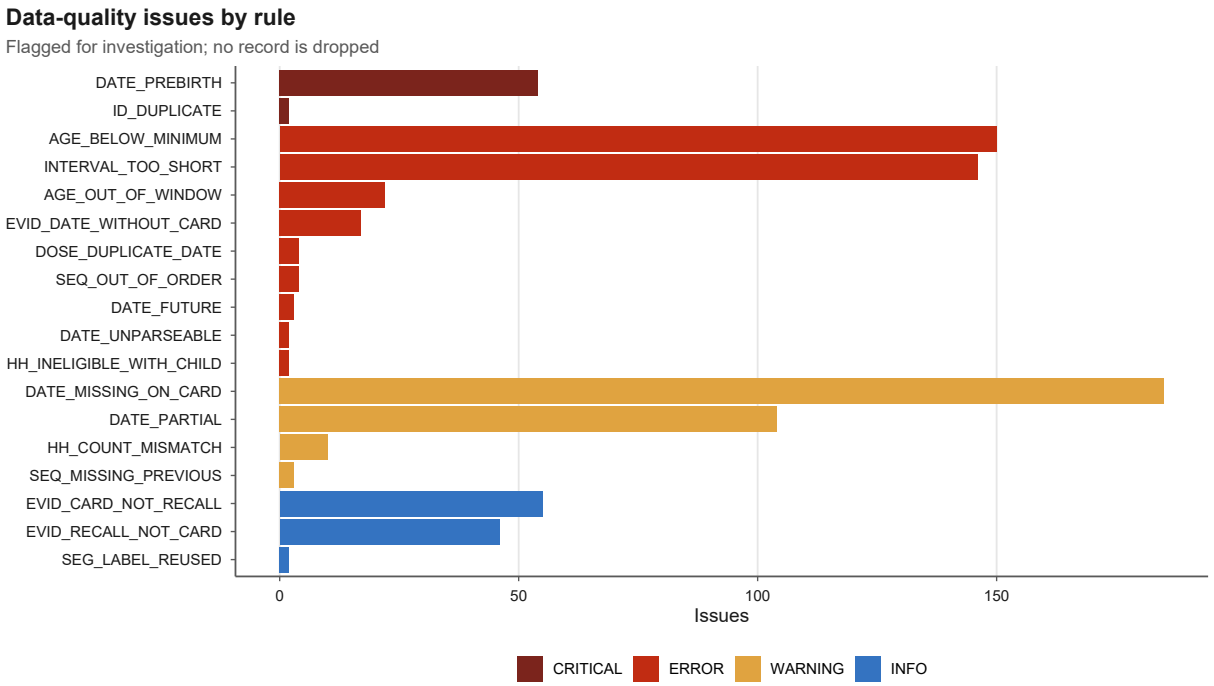
A child record is created wherever a repeat slot holds data, so empty slots in a wide export never become phantom children.

4 Check: flag, never silently correct

`validate_vcs()` returns an object, not a printed warning. Every issue carries a stable rule identifier, a severity and the record it concerns, so the result can be filtered, joined, exported or handed to the field team.

Table 5: Issues by severity: validate_vcs()

Severity	Issues	Records affected
CRITICAL	56	55
ERROR	350	304
WARNING	302	198
INFO	103	103



`clean_vcs()` then applies a ruleset. The default ruleset **flags and changes nothing**; any rule that does change a value writes a log line naming the record, the variable, the value before and after, the rule, its severity and the time.

Table 6: The audit trail: clean_vcs() then audit_changes()

Rule	Action	Severity	Records	Log entries
DATE_PREBIRTH	flag	CRITICAL	54	54
CARD_DATE_NO_CARD	flag	ERROR	17	17
DATE_FUTURE	flag	ERROR	3	3
DATE_UNPARSEABLE	flag	ERROR	2	2
DATE_PARTIAL	flag	WARNING	104	104

5 Derive: say what counts as vaccinated

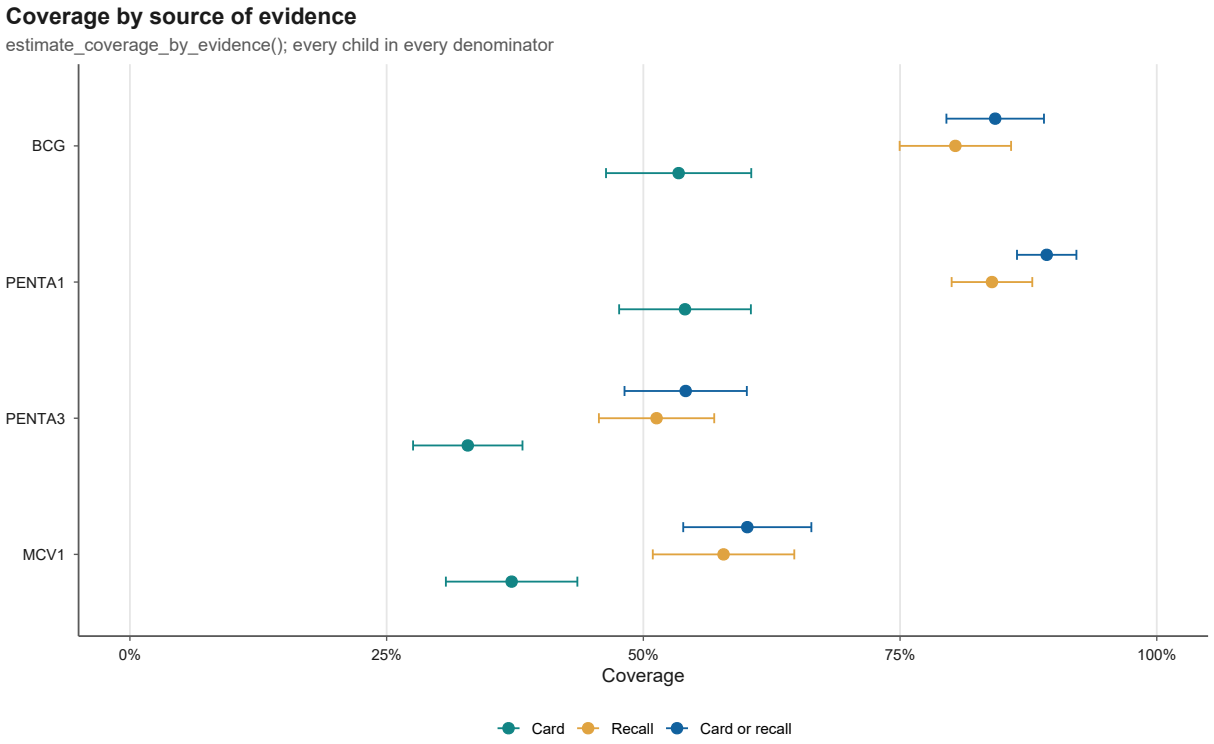
A coverage figure is meaningless without the definition behind it. The package grades the evidence for every dose and makes the definition an argument.

Table 7: Evidence grades: derive_evidence()

Evidence	Doses	Share
card_date	1161	32.4%
card_mark	185	5.2%
recall	931	26.0%
none	1270	35.4%
unknown	37	1.0%

`unknown` is not `none`. A dose with neither a card entry nor a recall answer is unknown, and `derive_vaccination_status()` leaves it `NA` rather than counting it as unvaccinated, unless you explicitly ask otherwise.

6 Estimate: keep the design to the end



The card row is *documented* coverage, not coverage among card holders: a child whose card was never seen counts as not documented, which is true. Pass `denominator = "determinable"` for the other convention, and say in the report which one a table uses.

Table 8: Two definitions of dropout, both reported: estimate_dropout()

Indicator	Method	n	Estimate	95% CI
PENTA1-PENTA3 dropout	individual	226	41.5%	35.8% – 47.2%
PENTA1-PENTA3 dropout	coverage	227	40.0%	33.5% – 46.4%

7 Tracking fieldwork against the frame

Progress is measured against the **sampling frame**, not the data: a cluster that was never visited leaves no trace in the export and can only come from the frame.

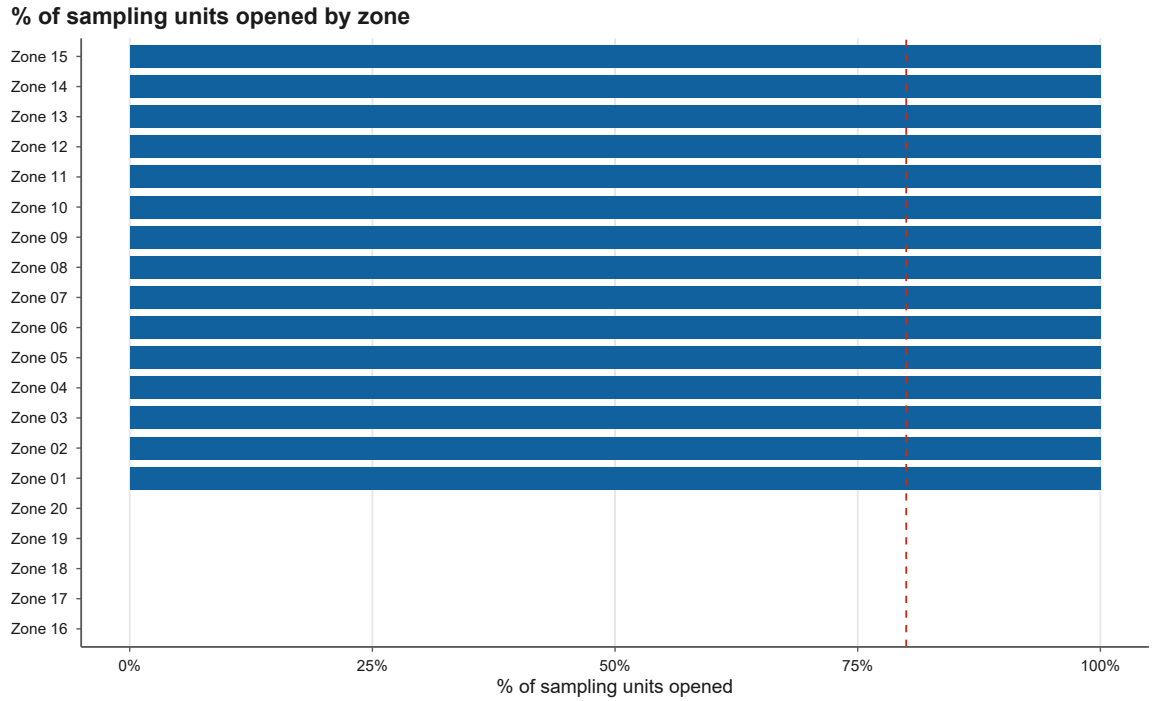


Table 9: The sampling-unit funnel: psu_map_summary()

Metric	N
Units in frame	40
Units opened	30
Units with children	30
Units with no eligible child	0
Children	256

Where a boundary layer is available, `plot_admin_map()` draws the same numbers as a choropleth with the sampling units bubbled on top, and units holding no data are drawn as "no data" rather than omitted.

8 Where to go next

Table 10: The seven vignettes, in reading order

Topic	Vignette
Getting started	vignette("vaxsurvR")
Mapping your own export	vignette("preparing-a-survey")
Quality checks	vignette("data-quality")
De-identification	vignette("deidentification")
Survey analysis	vignette("survey-analysis")
Coverage indicators	vignette("coverage-indicators")
End to end	vignette("end-to-end")

Produced with `vaxsurvR` 0.2.1 on 15 October 2026. Source: `inst/pipeline/vaxsurvR-pipeline.Rmd`. Rebuild with `rmarkdown::render(system.file("pipeline", "vaxsurvR-pipeline.Rmd", package = "vaxsurvR"))`.